

Academic Appointment

Dec. 2023– **Postdoctoral Fellow**, *Institute for Quantum Computing, University of Waterloo; Perimeter*

Institute for Theoretical Physics, Waterloo, Canada

Education

2017–2023 **Ph.D. in Physics**, *University of Illinois at Urbana-Champaign, Urbana, USA*

2013–2017 **B.Sc. in Physics**, *Nanjing University, Kuang Yaming Honor School, Nanjing, China*

Research Interests

My research seeks to understand, characterize, and certify genuinely nonclassical phenomena and harness them for information-processing and photonic applications.

Quantum foundations: Entanglement theory, quantum steering, Bell nonlocality, generalized noncontextuality, coherence, measurement incompatibility, and realist interpretations of quantum mechanics.

Optical quantum information: Quantum metrology, quantum communication, and distributed quantum information processing in quantum networks.

AI for Science (developing direction): Maintaining a website of open problems in quantum physics, translating loosely formulated physics questions into precise problems for LLM-assisted research, and exploring formal verification in Lean.

Honors and Awards

2023 **Boeing Quantum Creators Prize**. Chicago Quantum Exchange. Recognizes early-career researchers advancing quantum information science and engineering.

Preprints and Publications

7 preprints; 18 peer-reviewed journal articles.

13 first- or co-first-author papers (including 2 co-first-author papers and 3 preprints), with articles in PRX (2), PRL (1), and Quantum (3).

12 additional coauthored papers (including 4 preprints), with articles in PRL (1) and npj Quantum Information (1).

Preprints

- [P7] **Complexity of self-consistent entanglement certification**
Y. Zhang.
arXiv:2610.08757 (2026).
- [P6] **Destroying and Preserving Measurement Incompatibility over a Quantum Channel**
E. Chitambar and **Y. Zhang**.
arXiv:2610.06642 (2026).
- [P5] **Quantum key distribution using generalized contextuality against post-quantum eavesdroppers**
D. Centeno, R. D. Baldijão, M. Ciudad Alañón, **Y. Zhang**, P. Lauand, and E. Wolfe.
arXiv:2610.00595 (2026).
- [P4] **Compatibility of quantum instruments**
C. Kim, **Y. Zhang**, E. Chitambar, M. Junge, and F. Leditzky.
arXiv:2609.40266 (2026).
- [P3] **Exact Incompatibility-Breaking Criterion for Unital Qubit Channels**
Y. Zhang.
arXiv:2607.27757 (2026).
- [P2] **Quantum-Limited Subdiffraction Telescopy Requires Genuine Multi-Telescope Interference**
Y. Zhang, Y. Wang, W. Wu, and T. Jennewein.
arXiv:2606.27276 (2026).
- [P1] **Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark**
M. Zhu et al. (including **Y. Zhang**).
arXiv:2509.26574 (2025).

Peer-Reviewed Journal Articles

- [J18] **Entanglement Certification Using Noncontextuality Inequalities**
Y. Zhang, J. Spodek, D. Schmid, C. Reid, L. J. Morrison, T. Jennewein, K. J. Resch, and R. W. Spekkens.
Phys. Rev. X 16, 031057 (2026).
- [J17] **Quantifiers and witnesses for the nonclassicality of measurements and of states**
Y. Zhang, Y. Yīng, and D. Schmid.
Quantum 10, 2180 (2026).
- [J16] **Reassessing the Boundary between Classical and Nonclassical for Individual Quantum Processes**
Y. Zhang, D. Schmid, Y. Yīng, and R. W. Spekkens.
Phys. Rev. X 16, 021050 (2026).
- [J15] **Time-resolved certification of frequency-bin entanglement over multi-mode channels**
S. Vinet, M. Clementi, M. Bacchi, **Y. Zhang**, M. Giacomini, L. Neal, P. Villoresi, M. Galli, D. Bajoni, and T. Jennewein.
npj Quantum Information 12, 38 (2026).
- [J14] **Criteria for optimal entanglement-assisted long baseline telescopy**
Y. Zhang and T. Jennewein.
Phys. Rev. Research 7, 043278 (2025).

- [J13] **Cost of Simulating Entanglement in Steering Scenarios**
Y. Zhang, J. Zhang, and E. Chitambar.
Quantum 9, 1902 (2025).
- [J12] **Feasibility study of frequency-encoded photonic qubits over a free-space channel**
S. Vinet, W. Wu, Y. Zhang, and T. Jennewein.
Opt. Express 33, 40437-40449 (2025).
- [J11] **Temporally Localized Quantum Operations on Continuous-Wave Thermal Light**
Y. Wang, Y. Zhang, and V. O. Lorenz.
Phys. Rev. Lett. 135, 113602 (2025).
- [J10] **Astronomical interferometry using continuous-variable quantum teleportation**
Y. Wang, Y. Zhang, and V. O. Lorenz.
Phys. Rev. Research 7, 023154 (2025).
- [J9] **Public quantum network: The first node**
K. Kapoor, S. Hoseini, J. Choi, B. E. Nussbaum, Y. Zhang et al.
Appl. Phys. Lett. 126, 054002 (2025).
- [J8] **Exact Steering Bound for Two-Qubit Werner States**
Y. Zhang and E. Chitambar.
Phys. Rev. Lett. 132, 250201 (2024).
- [J7] **Information Carried by a Single Particle in Quantum Multiple-Access Channels**
X. Chen*, Y. Zhang*, A. Winter, V. O. Lorenz, and E. Chitambar.
Phys. Rev. A 109, 062420 (2024).
- [J6] **Fiber-based photon-pair generation: tutorial**
K. Garay-Palmett, D. B. Kim, Y. Zhang, F. Domínguez-Serna, V. O. Lorenz, and A. B. U'Ren.
J. Opt. Soc. Am. B 40, 469-490 (2023).
- [J5] **Building multiple access channels with a single particle**
Y. Zhang, X. Chen, and E. Chitambar.
Quantum 6, 653 (2022).
- [J4] **Superresolution in interferometric imaging of strong thermal sources**
Y. Wang, Y. Zhang, and V. O. Lorenz.
Phys. Rev. A 104, 022613 (2021).
- [J3] **Channel activation of CHSH nonlocality**
Y. Zhang*, R. A. Bravo*, V. O. Lorenz, and E. Chitambar.
New J. Phys. 22, 043003 (2020).
- [J2] **Dual-pump approach to photon-pair generation: demonstration of enhanced characterization and engineering capabilities**
Y. Zhang, R. Spiniolas, K. Shinbrough, B. Fang, O. Cohen, and V. O. Lorenz.
Opt. Express 27, 19050-19061 (2019).
- [J1] **Adaptive tomography of qubits: purity versus statistical fluctuations**
A. Zhang, Y. Zhang, F. Xu, L. Li, and L. Zhang.
J. Phys. A: Math. Theor. 51, 395304 (2018).

*Equal contribution (co-first authors).

Talks

- 2026 *A paradigm for entanglement certification using noncontextuality inequalities (invited)*. Foxconn.
- 2025 *Criteria for optimal entanglement-enhanced long baseline imaging protocols*. AQIS.
- Defining nonclassicality for individual quantum processes*. QPL.
- 2024 *Exact steering bound for two-qubit Werner states*. QIP.
- 2023 *Quantum-enhanced interferometric imaging (invited)*. Quantum 2.0.
- 2022 *Information carried by a single particle in multiple-access channels*. TQC.
- 2021 *Building multiple-access channels with one particle*. Beyond IID.
- 2020 *Dual-pump design enables novel photon-pair characterization and engineering*. CLEO.
- 2019 *Tailored photon-pair generation in optical fiber through dual-pump spontaneous four-wave mixing*. FiO-LS.

Outreach

- 2023 **Public Quantum Network at the Urbana Free Library**. Led the creation of a public quantum network distributing entangled photons from the University of Illinois to a local library, connecting quantum research with public education.
- 2023 **IQUIST: Ant-Man and the Wasp: Quantumania**. Led post-screening public discussions on quantum science at events coordinated by the Illinois Quantum Information Science and Technology Center (IQUIST).